The Chain Rule

ANSWER KEY

Section 1: Basic Chain Rule & Composite Functions

1 Differentiate $y = (3x^2 - 5x + 2)^4$.

a. $4(3x^2 - 5x + 2)^3(6x - 5)$ ✓

b. $4(6x - 5)^3$

c. $12x(3x^2 - 5x + 2)^3$

d. $4(3x^2 - 5x + 2)^3$

Chain rule $\frac{d}{dx}[u^4] = 4u^3 \cdot u'$. With $u = 3x^2 - 5x + 2$, $u' = 6x - 5$. Result is $4(3x^2 - 5x + 2)^3(6x - 5)$.

2 If $h(x) = f(g(x))$ with $g(3) = 5, g'(3) = -2,$ and $f'(5) = 7$, what is $h'(3)$?

a. -14 ✓

b. 10

c. -10

d. 35

By the chain rule: $h'(3) = f'(g(3)) \cdot g'(3) = f'(5) \cdot (-2) = 7 \cdot (-2) = -14$.

3 Find $\frac{d}{dx}[\cos(4x^3)]$.

a. $-12x^2\sin(4x^3)$ ✓

b. $-4\sin(4x^3)$

c. $12x^2\sin(4x^3)$

d. $-\sin(12x^2)$

Chain rule: $-\sin(4x^3) \cdot \frac{d}{dx}(4x^3) = -12x^2\sin(4x^3)$.

Section 2: Exponential & Logarithmic Chain Rule

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4 Find the derivative of $f(x) = e^{5x^2 - 2x}$.

a. $(10x - 2)e^{5x^2 - 2x}$ ✓

b. $e^{10x - 2}$

c. $10xe^{5x^2 - 2x}$

d. $(5x^2 - 2x)e^{5x^2 - 2x}$

$\frac{d}{dx}[e^u] = u'e^u$. Here $u' = 10x - 2$, so $f'(x) = (10x - 2)e^{5x^2 - 2x}$.

5 Differentiate $y = \ln(x^3 + 4)$.

a. $\frac{3x^2}{x^3 + 4}$ ✓

b. $\frac{1}{x^3 + 4}$

c. $\frac{3x^2}{x}$

d. $\frac{1}{3x^2}$

$\frac{d}{dx}[\ln u] = \frac{u'}{u}$. With $u = x^3 + 4$, $u' = 3x^2$, giving $\frac{3x^2}{x^3 + 4}$.

6 Find $\frac{d}{dx}[\ln(\cos x)]$.

a. $-\tan x$ ✓

b. $\tan x$

c. $\cot x$

d. $-\cot x$

$\frac{1}{\cos x} \cdot (-\sin x) = -\frac{\sin x}{\cos x} = -\tan x$.

Section 3: Repeated Chain Rule Applications

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7 Find the derivative of $y = \sin^3(2x)$.

a. $6\sin^2(2x)\cos(2x)$ ✓

b. $3\sin^2(2x)\cos(2x)$

c. $6\sin(2x)\cos(2x)$

d. $3\cos^2(2x)$

Outer power rule: $3\sin^2(2x)$. Middle trig rule: $\cos(2x)$. Inner linear rule: 2. Product $= 3 \cdot 2 \cdot \sin^2(2x)\cos(2x) = 6\sin^2(2x)\cos(2x)$.

8 Differentiate $f(x) = \sqrt{1 + \tan(3x)}$.

a. $\frac{3\sec^2(3x)}{2\sqrt{1+\tan(3x)}}$ ✓

b. $\frac{\sec^2(3x)}{2\sqrt{1+\tan(3x)}}$

c. $\frac{3\sec^2(3x)}{\sqrt{1+\tan(3x)}}$

d. $\frac{1}{2\sqrt{1+\tan(3x)}}$

$\frac{d}{dx}[\sqrt{u}] = \frac{u'}{2\sqrt{u}}$. Here $u' = 3\sec^2(3x)$.

9 If $g(x) = e^{\sin(4x)}$, what is $g'(0)$?

a. 4 ✓

b. 1

c. 0

d. $4e$

$g'(x) = 4\cos(4x)e^{\sin(4x)}$. At $x = 0$: $g'(0) = 4\cos(0)e^0 = 4(1)(1) = 4$.

Section 4: Table-Based Chain Rule Problems

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10 Let $h(x) = [f(x)]^3$. If $f(1) = 2$ and $f'(1) = -4$, what is $h'(1)$?

a. -48 ✓

b. -24

c. 12

d. -16

$h'(x) = 3[f(x)]^2 f'(x) \Rightarrow h'(1) = 3(2)^2(-4) = 3(4)(-4) = -48.$

11 Let $k(x) = f(x^2)$. If $f'(4) = 5$, what is $k'(2)$?

a. 20 ✓

b. 10

c. 5

d. 40

$k'(x) = f'(x^2) \cdot 2x \Rightarrow k'(2) = f'(4) \cdot 4 = 5 \cdot 4 = 20.$

12 If $F(x) = f(g(x))$ and $g(1) = 3, g'(1) = 2, f(3) = 5, f'(3) = 6$, find the equation of the tangent line to F at $x = 1$.

a. $y - 5 = 12(x - 1)$ ✓

b. $y - 3 = 12(x - 1)$

c. $y - 5 = 6(x - 1)$

d. $y - 5 = 2(x - 1)$

Point is $(1, F(1)) = (1, f(g(1))) = (1, f(3)) = (1, 5)$. Slope is $F'(1) = f'(g(1))g'(1) = f'(3) \cdot 2 = 6 \cdot 2 = 12$. Tangent line: $y - 5 = 12(x - 1)$.

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